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# **GCSE MARKING SCHEME**

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**SUMMER 2017**

**GCSE (NEW)**

**APPLIED SCIENCE (DOUBLE AWARD) - UNIT 2**

**3445U20-1 / 3445UB0-1**

## **INTRODUCTION**

This marking scheme was used by WJEC for the 2017 examination. It was finalised after detailed discussion at examiners' conferences by all the examiners involved in the assessment. The conference was held shortly after the paper was taken so that reference could be made to the full range of candidates' responses, with photocopied scripts forming the basis of discussion. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners.

It is hoped that this information will be of assistance to centres but it is recognised at the same time that, without the benefit of participation in the examiners' conference, teachers may have different views on certain matters of detail or interpretation.

WJEC regrets that it cannot enter into any discussion or correspondence about this marking scheme.

**GCSE APPLIED SCIENCE (DOUBLE) AWARD**

**UNIT 2 (NEW) 3445U20-1/3445UB0-1**

**SUMMER 2017 MARK SCHEME**

| Question |     |       |  | Marking details                                                                                                      | Marks Available |     |     |       |       |      |
|----------|-----|-------|--|----------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                      | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 1<br>FT  | (a) | (i)   |  | oil will last longer/won't use as much oil                                                                           | 1               |     |     |       |       |      |
|          |     | (ii)  |  | any 1 × (1) from:<br>more landfill needed (1)<br>take a long time to biodegrade / rot (1)                            | 1               |     |     |       |       |      |
|          |     | (iii) |  | any 2 × (1) from:<br>animals could suffocate (1)<br>animals could get tangled in bags (1)<br>animals may swallow (1) | 2               |     |     |       |       |      |
|          | (b) | (i)   |  | 280 [million]                                                                                                        |                 | 1   |     |       | 1     |      |
|          |     | (ii)  |  | $\frac{70}{350} \times 100$ (1)<br>= 20% (1)<br><br>Answer only (2)                                                  |                 | 2   |     |       | 2     |      |
|          | (c) | (i)   |  | pathogens                                                                                                            | 1               |     |     |       |       |      |
|          |     | (ii)  |  | White blood cells                                                                                                    | 1               |     |     |       |       |      |
|          |     |       |  | Question 1 total                                                                                                     | 6               | 3   | 0   | 9     | 3     | 0    |

| Question |     |      |  | Marking details                                                                                                                                                   | Marks Available |          |          |          |          |          |
|----------|-----|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                   | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 2<br>FT  | (a) | (i)  |  | H                                                                                                                                                                 | 1               |          |          |          |          |          |
|          |     | (ii) |  | Hh or HH                                                                                                                                                          | 1               |          |          |          |          |          |
|          | (b) |      |  | <p>(2 × 1)</p> <p>50% chance (1) compared with 75% chance (1)</p>                                                                                                 |                 | 4        |          |          |          |          |
|          | (c) |      |  | <p>whether to have children / whether to terminate pregnancy / risk of foetal testing / possible misuse of information</p> <p>Accept: may pass on to children</p> | 1               |          |          |          |          |          |
|          |     |      |  | <b>Question 2 total</b>                                                                                                                                           | <b>3</b>        | <b>4</b> | <b>0</b> | <b>7</b> | <b>0</b> | <b>0</b> |

| Question       |     |       |  | Marking details                                                 | Marks Available |          |          |          |          |          |
|----------------|-----|-------|--|-----------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|                |     |       |  |                                                                 | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| <b>3</b><br>FT | (a) |       |  | Absorbed (1)<br>Chemical make up (1)<br>Moving away from us (1) | 3               |          |          |          |          |          |
|                | (b) | (i)   |  | 1                                                               |                 | 1        |          |          |          |          |
|                |     | (ii)  |  | 3                                                               |                 | 1        |          |          |          |          |
|                |     | (iii) |  | The same                                                        |                 |          | 1        |          |          |          |
|                |     | (iv)  |  | 1                                                               |                 | 1        |          |          |          |          |
|                |     |       |  | <b>Question 3 total</b>                                         | <b>3</b>        | <b>3</b> | <b>1</b> | <b>7</b> | <b>0</b> | <b>0</b> |

| Question |     |       |  | Marking details                                                                                                                                                   | Marks Available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4<br>FT  | (a) | (i)   |  | 1 in 6 / $\frac{1}{6}$ / $\frac{20}{120}$                                                                                                                         |                 | 1   |     |       | 1     | 1    |
|          |     | (ii)  |  | 20                                                                                                                                                                |                 |     | 1   |       |       | 1    |
|          |     | (iii) |  | Remove those with a 6 (or any other number) facing up                                                                                                             | 1               |     |     |       |       | 1    |
|          |     | (iv)  |  | The remaining dice were rolled again (and again) (1)<br>taking away the 6's (or stated side) each time (1)                                                        | 2               |     |     |       |       | 2    |
|          | (b) | (i)   |  | 19 (1)<br>50 (1)                                                                                                                                                  |                 | 2   |     |       |       | 2    |
|          |     | (ii)  |  | 60                                                                                                                                                                |                 | 1   |     |       | 1     | 1    |
|          |     | (iii) |  | 4 throws                                                                                                                                                          |                 |     | 1   |       | 1     | 1    |
|          |     | (iv)  |  | 120 → 60 → 30 → 15 (1)<br>Half-lives =3 (1)<br><br>Accept 3 on answer line for 2 marks                                                                            |                 | 2   |     |       | 2     | 2    |
|          | (c) | (i)   |  | Longer <u>half-life</u> (1)<br>So will not need replacing as often (1)<br><br>OR<br><br>Chromium has a shorter half-life (1)<br>So needs replacing more often (1) |                 |     | 2   |       |       |      |
|          |     | (ii)  |  | Beta and gamma (emitter) (1)<br>will penetrate packing (1)<br><br>OR<br><br>Bismuth emits alpha (1)<br>Which won't penetrate the packing (1)                      |                 |     | 2   |       |       |      |
|          |     |       |  | Question 4 total                                                                                                                                                  | 3               | 6   | 6   | 15    | 5     | 11   |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks Available |     |     |       |       |      |
|----------|-----|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5<br>FT  | (a) |  |  | <p><b>Indicative content</b><br/>Blood glucose levels are raised after mealtimes, pancreas secretes insulin, liver converts glucose to <u>glycogen</u>, which is stored in the liver and blood <u>glucose</u> returns to normal.</p> <p>If blood glucose is too low, the pancreas secretes <u>glucagon</u>, and liver convert glycogen into glucose and blood glucose returns to normal.</p> <p><b>5-6 marks</b><br/>The candidate correctly describes link between changes in glucose concentration and insulin and glucagon release. Role of pancreas, liver correctly described.<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3-4 marks</b><br/>The candidate partly described the link between insulin or glucagon release and glucose concentration. They correctly describe the role of the pancreas, liver.<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> | 6               |     |     |       |       |      |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks Available |     |     |           |       |          |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-----------|-------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total     | Maths | Prac     |
|          |     |      |  | <b>1-2 marks</b><br>The candidate describes that glucose levels rise after a meal and decreases at other times.<br><br><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i><br><br><b>0 marks</b><br>No attempt made or no response worthy of credit. |                 |     |     |           |       |          |
|          | (b) |      |  | any 2 × (1) from:<br>control/low sugar diet (1)<br><u>insulin</u> injections (1)<br>pancreatic tissue transplant (1)                                                                                                                                                                                                                                                                                                                                         | 2               |     |     |           |       |          |
|          | (c) | (i)  |  | Benedict's test (1)<br>Boil/heat (1)                                                                                                                                                                                                                                                                                                                                                                                                                         | 2               |     |     |           |       |          |
|          |     | (ii) |  | (blue→) green / amber / (brick) red (1)<br>Blue/no colour change (1)                                                                                                                                                                                                                                                                                                                                                                                         | 2               |     |     |           |       |          |
|          |     |      |  | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>12</b>       |     |     | <b>12</b> |       | <b>6</b> |

| Question           |     |       |  | Marking details                                                                                                                                                                                                                                                                               | Marks Available |     |     |       |       |      |
|--------------------|-----|-------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|                    |     |       |  |                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6<br>FT            | (a) | (i)   |  | 92.1                                                                                                                                                                                                                                                                                          | 1               |     |     |       |       |      |
|                    |     | (ii)  |  | 1995                                                                                                                                                                                                                                                                                          | 1               |     |     |       |       |      |
|                    | (b) | (i)   |  | 897                                                                                                                                                                                                                                                                                           |                 | 1   |     |       |       |      |
|                    |     | (ii)  |  | CFC                                                                                                                                                                                                                                                                                           | 1               |     |     |       |       |      |
|                    | (c) |       |  | Temperature changes track sunspot changes up to {(about)1940/initially/early on the time scale} (1)<br>Temperature trend tracks CO <sub>2</sub> after {(about) 1950/after on the timescale}(1)<br><br>Accept: the trend tracks CO <sub>2</sub> all along (1)                                  |                 |     | 2   |       | 2     |      |
| 7<br>FT<br>1<br>HT | (a) | (i)   |  | Any 2 × (1) from:<br>Increases after 1990 or by 1995 (1)<br>Drops in 2000 (1)<br>Rises by 2005/2010 (1)<br>Remains approximately steady after 2010 (1)<br><br>Each marking point must include a relevant year<br><br>Increases, decreases and increases (1 mark only)                         |                 | 2   |     |       | 3     |      |
|                    |     | (ii)  |  | Scales (x axis 5 years per 2 cm square, y axis 50 per 2 cm square) (1)<br><br>7 correct plots (within appropriate small square) (2)<br>6 correct plots (within appropriate small square) (1)<br>5 correct plots (0)<br><br>curve of best fit / joining points with curve / point to point (1) |                 | 4   |     |       | 4     |      |
|                    |     | (iii) |  | CO <sub>2</sub> emissions decrease (over time)/negative correlation (1)<br>Rate of decrease in the last 10 years is greater than the first 10 years / smaller decrease before 2005 than after 2005 (1)<br><br>Ignore reference to plateau. Allow ecf from incorrect plot                      |                 | 2   |     |       |       |      |

| Question |  |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks Available |     |     |       |       |      |
|----------|--|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  | (iv) |  | Graph 1 shows an increase over time (1)<br>so does not correlate with drawn graph (1)<br><br>OR<br>My graph shows emissions (from UK) decrease (1)<br>but atmospheric concentration is increasing (1)                                                                                                                                                                                                                                                      |                 |     | 2   |       |       |      |
|          |  | (v)  |  | Rate of photosynthesis varies throughout the year (1)<br>Uptake of CO <sub>2</sub> varies (1)<br>Lower rate (in November) corresponds with peak / Higher rate (in May) correlates with dip (1)<br><br>Accept:<br>Rate of decay varies throughout the year (1)<br>CO <sub>2</sub> release varies (1)<br>higher rate (of decay/respiration of decomposers) corresponds with peak / lower rate (of decay/respiration of decomposers) corresponds with dip (1) |                 |     | 3   |       |       |      |

| Question |                   |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |        |         |                           | Marks Available |     |     |       |       |      |
|----------|-------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|---------|---------------------------|-----------------|-----|-----|-------|-------|------|
|          |                   |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |        |         |                           | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b)               |  |  | 288 000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 68 520 | 11 880 | 368 400 | 72.37                     |                 |     |     |       |       | (b)  |
|          |                   |  |  | 17 808                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 12 117 | 6 720  | 36 645  | 7.20/7.1986/<br>7.19/7.00 |                 | 3   |     |       | 3     |      |
|          |                   |  |  | 3 × (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |        |         |                           |                 |     |     |       |       |      |
|          | (c)               |  |  | Water vapour becomes largest factor (1)<br>All others decrease (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |        |         |                           |                 | 2   |     |       |       | (c)  |
|          | (d)               |  |  | man made contributions are very small / natural contributions are high                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |        |         |                           |                 |     | 1   |       |       | (d)  |
|          | (e)<br>HT<br>Only |  |  | <p><b>Indicative content</b></p> <p>In graph 2, variations in the sunspot cycle length and the temperature anomaly are closely linked. This is true for the whole time scale. Between 1910 and 1970 the CO<sub>2</sub> concentration gradually increases but during this time the temperature anomaly increases steeply, peaks and falls.</p> <p>In graph 3, there is a close match between the increasing trends of CO<sub>2</sub> concentration and temperature anomaly. The steepest increase in temperature since 1960 is matched by a fluctuating but steady sunspot activity.</p> <p>The evidence is not conclusive since graph 2 supports solar activity being responsible for global warming but graph 3 supports the CO<sub>2</sub> cause.</p> <p><b>5-6 marks</b></p> <p>Comprehensive comparison of trends of the 3 variables in both graphs. A statement that the evidence is inconclusive is required.</p> <p><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> |        |        |         |                           |                 |     |     |       |       |      |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks Available |           |          |           |           |          |
|----------|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|----------|-----------|-----------|----------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2       | AO3      | Total     | Maths     | Prac     |
|          |  |  |  | <p><b>3-4 marks</b><br/>Comparison of trends of at least two variables in both graphs. May not state that evidence in graphs contradicts each other. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> <p><b>1-2 marks</b><br/>Limited comparisons made of trends or some analysis of one graph. No conclusions made. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/>No attempt made or no response worthy of credit.</p> |                 |           |          |           |           |          |
|          |  |  |  | <b>Section B total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>3</b>        | <b>14</b> | <b>8</b> | <b>25</b> | <b>12</b> | <b>0</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                | Marks Available |          |          |           |          |           |
|----------|-----|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|-----------|
|          |     |       |  |                                                                                                                                                                                                                                                                                | AO1             | AO2      | AO3      | Total     | Maths    | Prac      |
| 2<br>HT  | (a) | (i)   |  | <ul style="list-style-type: none"> <li>The 90 dice were rolled (1)</li> <li>Any that landed with a 6 (or any other) facing upwards were removed (1)</li> <li>The remaining dice were rolled again and again, taking away the 6's (or the same number) each time (1)</li> </ul> | 3               |          |          |           |          | 3         |
|          |     | (ii)  |  | 15                                                                                                                                                                                                                                                                             |                 | 1        |          |           | 1        | 1         |
|          |     | (iii) |  | smooths out variations/finds anomalies/more repeatable/ more reproducible                                                                                                                                                                                                      | 1               |          |          |           |          | 1         |
|          | (b) | (i)   |  | 85<br>185                                                                                                                                                                                                                                                                      |                 | 2        |          |           |          | 2         |
|          |     | (ii)  |  | 4 throws                                                                                                                                                                                                                                                                       |                 | 1        |          |           | 1        | 1         |
|          | (c) | (i)   |  | Any construction lines (1)<br>Around 3.8-3.9 (1)                                                                                                                                                                                                                               |                 | 2        |          |           | 2        | 2         |
|          |     | (ii)  |  | More accurate                                                                                                                                                                                                                                                                  | 1               |          |          |           |          | 1         |
|          | (d) | (i)   |  | Caesium-137                                                                                                                                                                                                                                                                    |                 |          | 1        |           |          |           |
|          |     | (ii)  |  | Beta and gamma will penetrate packing (1)<br>Will not need replacing very often due to its half-life (1)<br><br>If chromium selected in d(i)<br>Accept: gamma will penetrate packing (1)                                                                                       |                 |          | 2        |           |          |           |
|          |     |       |  | <b>Question 2 total</b>                                                                                                                                                                                                                                                        | <b>5</b>        | <b>6</b> | <b>3</b> | <b>14</b> | <b>4</b> | <b>11</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks Available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3<br>HT  | (a) | (i)   |  | A form of a gene                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1               |     |     |       |       |      |
|          |     | (ii)  |  | Masks the contribution of another allele                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1               |     |     |       |       |      |
|          |     | (iii) |  | Genetic make up of a pair of alleles/type of genes an individual has                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1               |     |     |       |       |      |
|          | (b) |       |  | <div><div><div>Parent with Huntington's</div><div><div>H</div><div>h</div></div><div><div>Normal Parent</div><div><div>h</div><div>h</div></div><div><div>Hh</div><div>Hh</div><div>hh</div><div>hh</div></div></div><div><div>Parent with Huntington's</div><div><div>H</div><div>h</div></div><div><div>Parent with Huntington's</div><div><div>H</div><div>h</div></div><div><div>Hh</div><div>Hh</div><div>Hh</div><div>hh</div></div></div></div><p>(Gametes (1) cross (1) (ecf) ) × 2</p><p>50% chance compared with 75% chance of being affected (ecf)(1)</p><p>Accept 50% chance compared to 25% chance of not being affected (ecf)</p></div></div> |                 | 5   |     |       | 1     |      |
|          | (c) |       |  | Any 2 × (1) from:<br>whether to have children (1)<br>whether to terminate pregnancy (1)<br>risk of foetal testing (1)<br>possible misuse of information (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2               |     |     |       |       |      |
|          |     |       |  | Question 3 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5               | 5   |     | 10    | 1     |      |

| Question |  |       |  | Marking details                                                                                                                                                                                            | Marks Available |     |     |           |       |          |
|----------|--|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-----------|-------|----------|
|          |  |       |  |                                                                                                                                                                                                            | AO1             | AO2 | AO3 | Total     | Maths | Prac     |
| 4<br>HT  |  | (i)   |  | A Sensory neuron/sensory nerve (1)<br>B Spinal cord (1)<br>C Motor neuron/motor nerve (1)                                                                                                                  | 3               |     |     |           |       |          |
|          |  | (ii)  |  | Impulse sent along <u>sensory</u> neuron (1)<br>Travels through <u>relay</u> neuron (1)<br>Sent back to muscle through <u>motor</u> neuron (1)<br>Causes extensor muscle to contract (1)                   | 4               |     |     |           |       | 4        |
|          |  | (iii) |  | Extensor muscle <u>contracts</u> flexor <u>relaxes</u> (1)<br>Leg straightens (1 or*)<br>Flexor <u>contracts</u> extensor <u>relaxes</u> (1)<br>Leg bends (or 1*)<br><br>Accept either* in correct context | 3               |     |     |           |       |          |
|          |  |       |  | <b>Question 4 total</b>                                                                                                                                                                                    | <b>10</b>       |     |     | <b>10</b> |       | <b>4</b> |

| Question |     |      |  | Marking details                                                                                                                                                                    | Marks Available |     |     |       |       |      |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                    | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5<br>HT  | (a) | (i)  |  | Natural resources will last longer (1)<br>Used for other purposes (1)                                                                                                              | 2               |     |     |       |       |      |
|          |     | (ii) |  | Any 3 × (1) from:<br>Less litter/landfill/waste (1)<br>Less harm to wildlife (1)<br>Less damage caused by extracting natural resources (1)<br>Less emissions during production (1) | 3               |     |     |       |       |      |
|          | (b) |      |  | Dropped to 1/5 or 20% (1)<br>So assume similar drop in England of 20% (1)<br>= 1.5 billion / 1500 million (1)<br><br>Accept 1 500 million for (3)<br><br>1500 only award (2)       |                 | 3   |     |       | 1     |      |
|          | (c) |      |  | Some bacteria are harmful/pathogens/cause disease/make you ill (1)<br>Cause <u>food poisoning</u> in humans (1)                                                                    | 2               |     |     |       |       |      |
|          |     |      |  | Question 5 total                                                                                                                                                                   | 7               | 3   |     | 10    | 1     |      |

| Question |     |  |  | Marking details                                                                                                                  | Marks Available |          |     |          |       |      |
|----------|-----|--|--|----------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|-----|----------|-------|------|
|          |     |  |  |                                                                                                                                  | AO1             | AO2      | AO3 | Total    | Maths | Prac |
| 6<br>HT  | (a) |  |  | Absorption of light as it passes through (named) gas (1)<br>Characteristic of elements present (1)<br>At certain wavelengths (1) | 3               |          |     |          |       |      |
|          | (b) |  |  | Same <u>pattern</u> of lines (1)<br>All <u>red-shifted</u> (1)<br>By different amounts/more/less (1)                             |                 | 3        |     |          |       |      |
|          |     |  |  | <b>Question 6 total</b>                                                                                                          | <b>3</b>        | <b>3</b> |     | <b>6</b> |       |      |

**SUMMARY HT**

| Question     | AO1       | AO2       | AO3       | TOTAL MARK | MATHS     | PRAC      |
|--------------|-----------|-----------|-----------|------------|-----------|-----------|
| 1            | 0         | 13        | 12        | 25         | 10        | 0         |
| 2            | 5         | 6         | 3         | 14         | 4         | 11        |
| 3            | 5         | 5         | 0         | 10         | 1         | 0         |
| 4            | 10        | 0         | 0         | 10         | 0         | 4         |
| 5            | 7         | 3         | 0         | 10         | 1         | 0         |
| 6            | 3         | 3         | 0         | 6          | 0         | 0         |
| <b>TOTAL</b> | <b>30</b> | <b>30</b> | <b>15</b> | <b>75</b>  | <b>16</b> | <b>15</b> |

**SUMMARY FT**

| Question     | AO1       | AO2       | AO3       | TOTAL MARK | MATHS     | PRAC      |
|--------------|-----------|-----------|-----------|------------|-----------|-----------|
| 1            | 6         | 3         | 0         | 9          | 3         | 0         |
| 2            | 3         | 4         | 0         | 7          | 0         | 0         |
| 3            | 3         | 3         | 1         | 7          | 0         | 0         |
| 4            | 3         | 6         | 6         | 15         | 5         | 11        |
| 5            | 12        | 0         | 0         | 12         | 0         | 4         |
| 6+7          | 3         | 14        | 8         | 25         | 12        | 0         |
| <b>TOTAL</b> | <b>30</b> | <b>30</b> | <b>15</b> | <b>75</b>  | <b>20</b> | <b>15</b> |